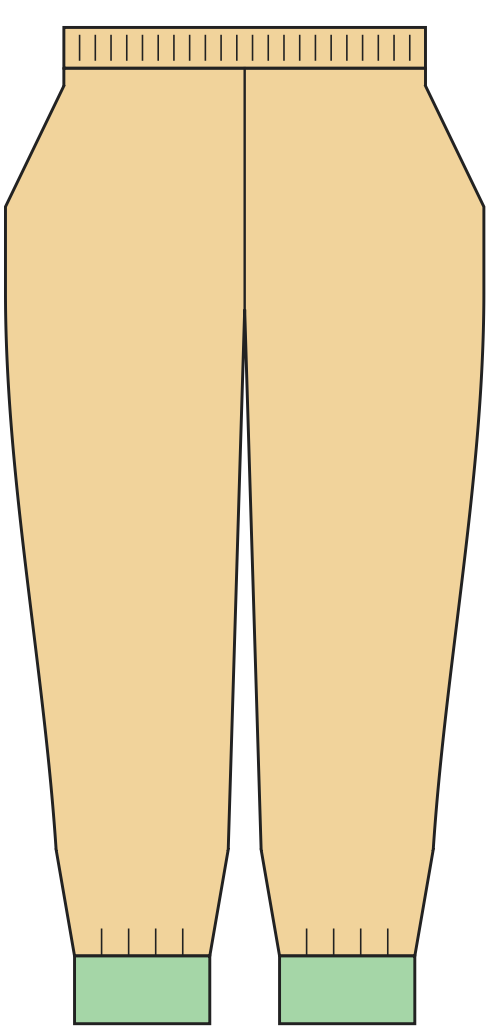
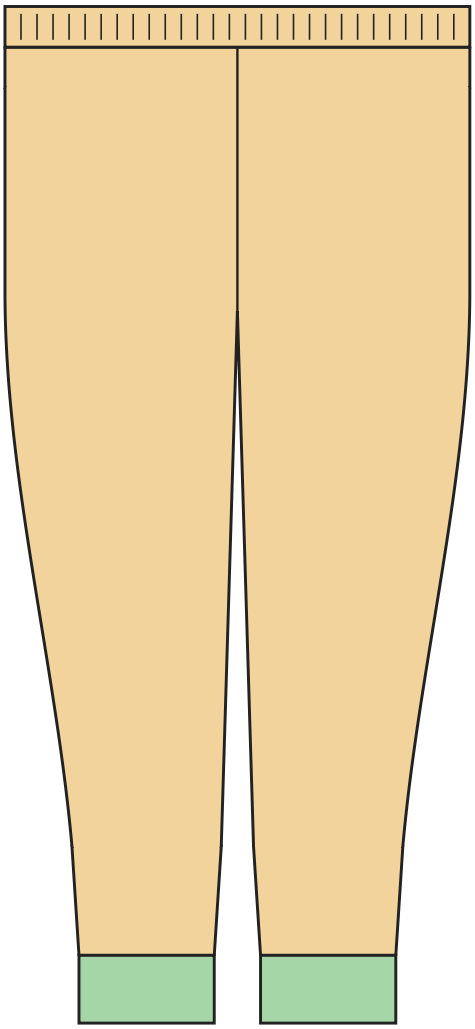
# Joggers

size 122 · normal fit, Sweat / French Terry (25 % stretch)

6 parts · 25 steps · medium · about 3.5 h



Front view



Back view

**Sewn with: Sewing machine**



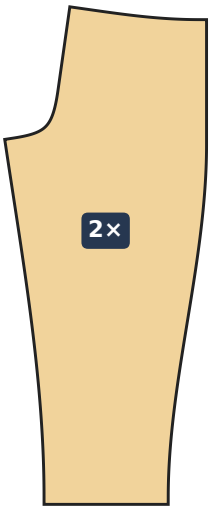
Sewing machine



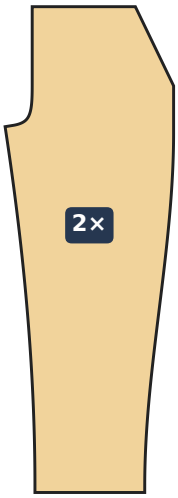
Iron

Read the steps once before you start. Every picture is drawn from your pattern pieces. Sew a test garment from cheap fabric first and adjust before you cut the good fabric.

# Parts




B · Back leg




A · Front leg




C · Side panel (outer pocket bag)




D · Inner pocket bag




F · Leg cuff




E · Waistband (elastic) (cut on fold)

Seam allowance: included: 1.0 cm · hem 1.0 cm

# Material

**Main fabric**
0.90 m at 150 cm width

**Rib knit**
0.20 m at 70 cm width

**Elastic**
Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

**Sewing machine**
narrow zigzag 1 × 2.5 mm (or overlocker); twin needle 4 mm, stitch 3 mm; longest straight stitch, no backstitch; wide zigzag 4 × 2.5 mm

**Iron**
steam

**Polyester sewing thread**
In the colour of the fabric.

**Pins or clips**
Pin across the seam line.

**Chalk or a fabric pen**
For notches, labels and placement lines.

## Colours in this booklet

**Main fabric**
right side · wrong side

**Rib knit**
right side · wrong side

Sew now (machine)

Sew by hand

Already sewn

Fold / press line

Marking (chalk)

Pin

Notch: match them

Right/wrong side of the fabric = the side of the fabric. Left/right (as worn) = seen from the child wearing it: the left front leg is on the right when you look at the child.

# Basics: Sewing machine

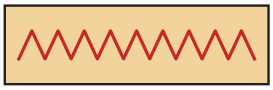
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

## Basics: the sewing machine



### Straight stitch

Seams in woven fabric, 2.5 mm; backstitch at the start and the end.



### Zigzag

Knit seams (narrow) and finishing the edges (wide).

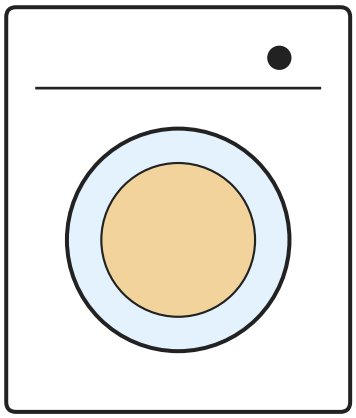


### Topstitching

Right side up, with the same distance to the edge all the way; use the edge of the presser foot as a guide.

## 1 Pre-wash the fabric

1. Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.



30–40°

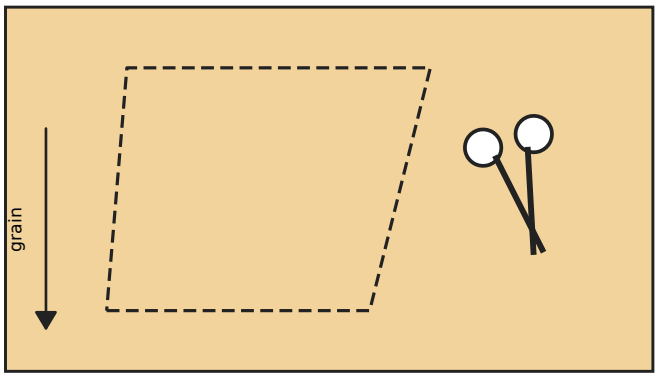
## 2 Cut out all pieces



Scissors

1. Follow the cutting layout
2. Grainline arrows parallel to the selvage.

Tip: Cut both layers at once with long strokes; keep the fabric flat on the table.

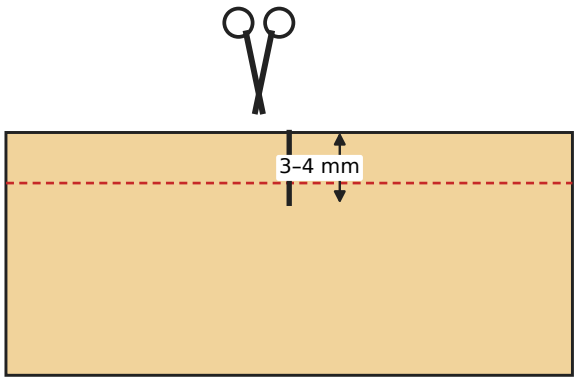


## 3 Snip the notches

Chalk

1. Cut 3–4 mm into the seam allowance at every notch.

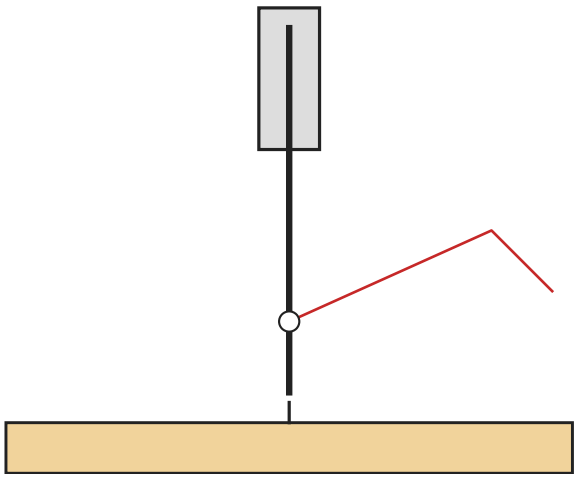
⚠ Important: Never deeper than half the seam allowance.




## 4 Set up the machine

1. Jersey/stretch needle 75–80, polyester thread.
2. Sew a test seam on a scrap first.

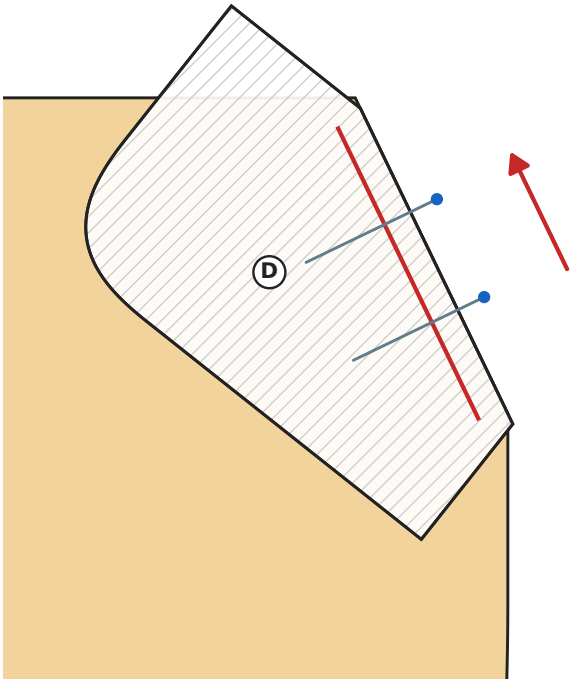
Tip: See the basics page at the front of the booklet.



## 5 Sew the pocket bag to the opening

Sewing machine

Pins




1. Lay the inner pocket bag right sides together on the pocket opening of the front leg, edges level, and pin.
2. Sew along the opening with 10 mm seam allowance.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: The opening is on the bias: do not stretch it.

Tip: A strip of stay tape ironed onto the opening keeps it in shape.

2×

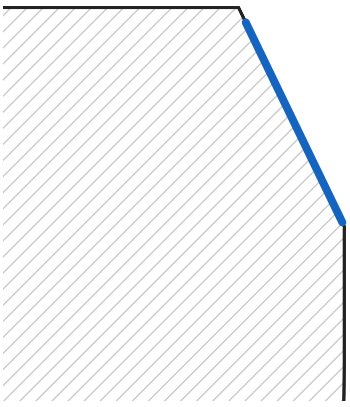
## 6 Clip and turn the bag inside

Iron

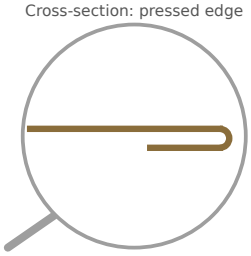
1. Clip the curved allowance every 1 cm, close to the stitching.
2. Turn the bag to the inside and press the edge.

Iron: steam

⚠ Important: Roll the edge 1–2 mm to the inside: no pocket bag may show from the outside.



Cross-section: pressed edge



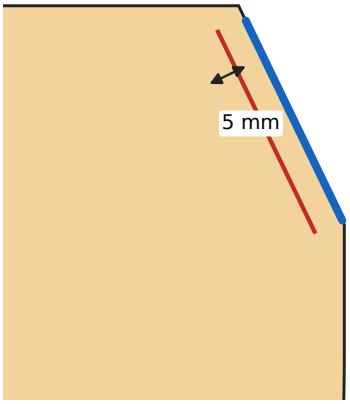
2×

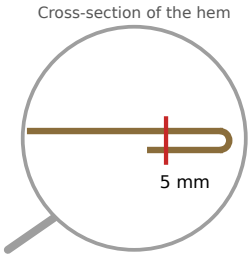
## 7 Topstitch the pocket edge

Sewing machine

1. Right side up, topstitch 5 mm from the edge through all layers.

Sewing machine: twin needle 4 mm, stitch 3 mm

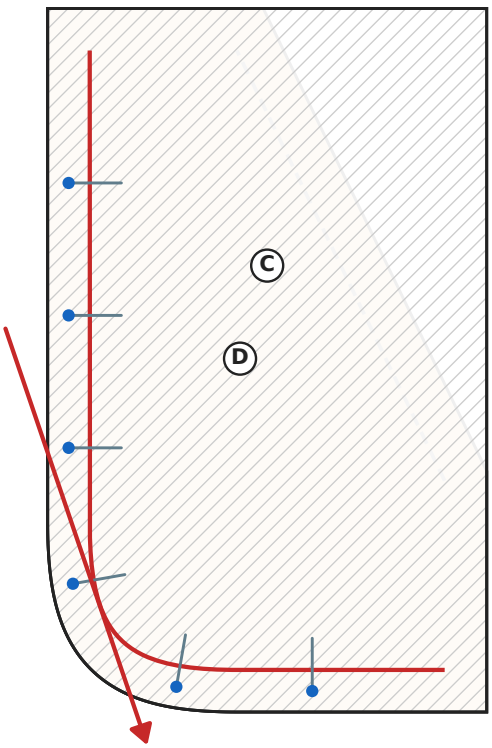



2×

## 8 Sew the pocket bag seam

Sewing machine · Pins




1. Lay the side panel right sides together under the pocket bag so that the "pocket opening" mark meets the opening edge.
2. Pin the two pocket bag layers and sew them together along the curve with 10 mm seam allowance.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

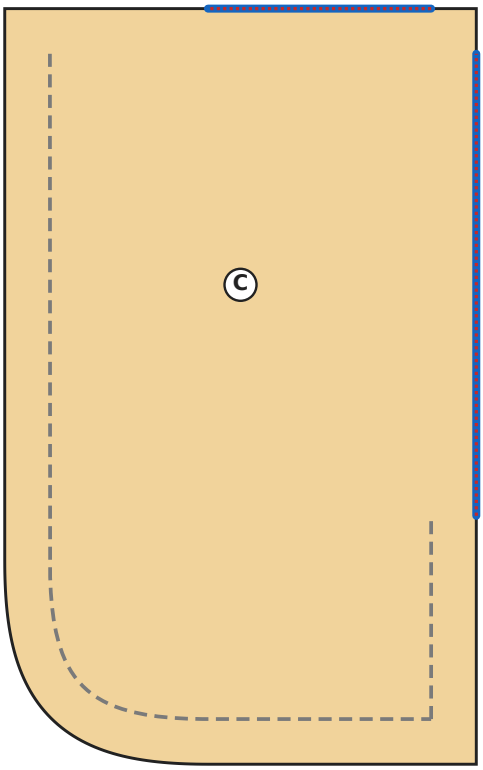
⚠ Important: Fold the front leg out of the way: sew only the two pocket bag layers.

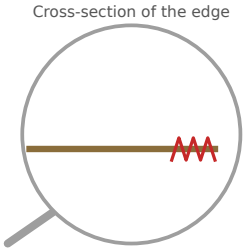
Tip: Turn it over afterwards and check from the outside that the opening lies flat.

2×

## 9 Baste the side panel to the leg

Sewing machine




1. Lay the leg flat on the side panel.
2. Baste the top edge and the side edge together, inside the seam allowance.

Sewing machine: longest straight stitch, no backstitch

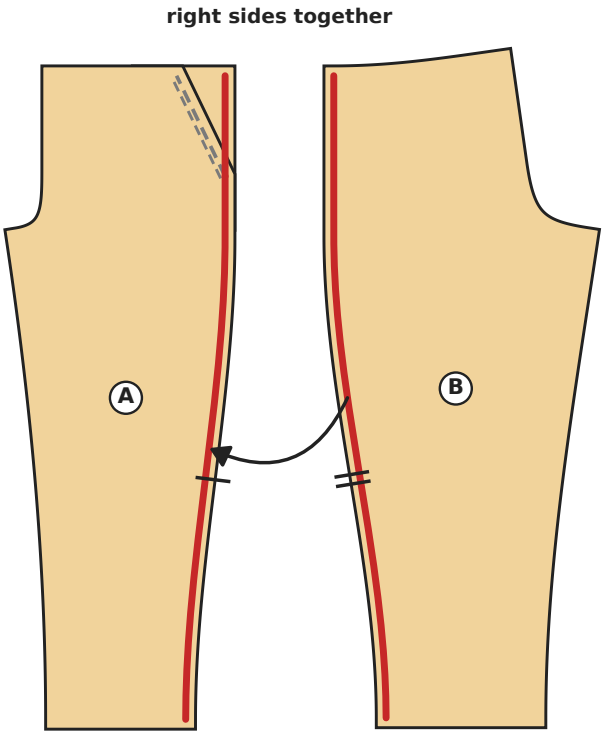
⚠ Important: The pocket opening lies flat, not pulled.

Tip: Baste 4 mm from the edge, inside the seam allowance.

2×

## 10 Sew the side seams

Sewing machine  Pins




1. Lay front and back of the same leg right sides together, side edges level: single notches = front, double notches = back.
2. Pin, then sew with 10 mm seam allowance from the top edge to the hem.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

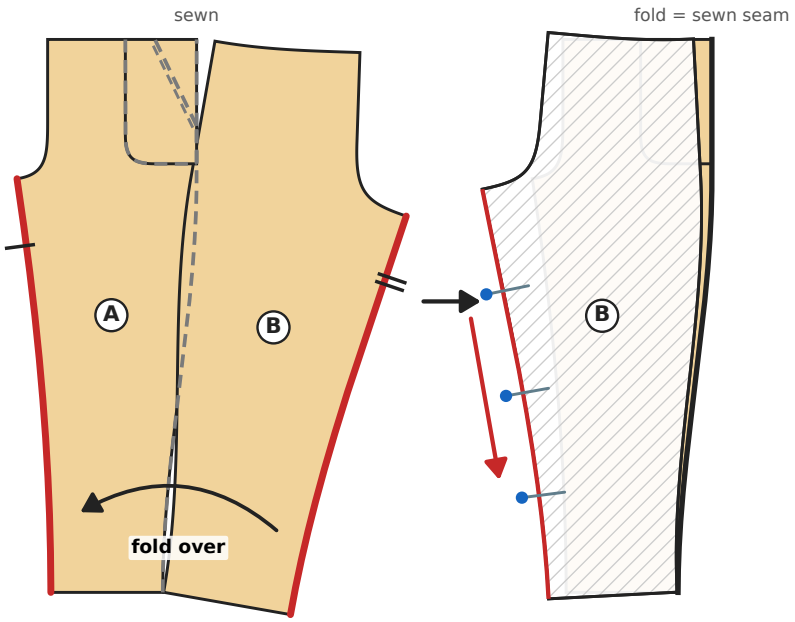
⚠ Important: Always the front and back of the SAME leg (FL + BL, FR + BR).

2×

## 11 Sew the inseams

Sewing machine

Pins



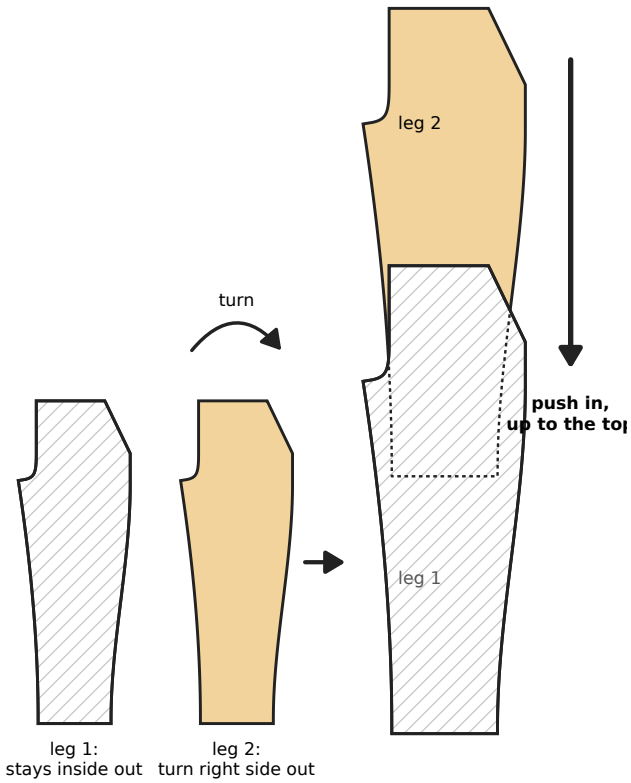

1. Open the leg out at the side seam and fold it right sides together, inseam edges level.
2. Pin the notches first, then the ends.
3. Sew with 10 mm seam allowance from the crotch point to the hem.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Start exactly at the crotch point: the four seams meet there later.

2×

## 12 Put one leg inside the other

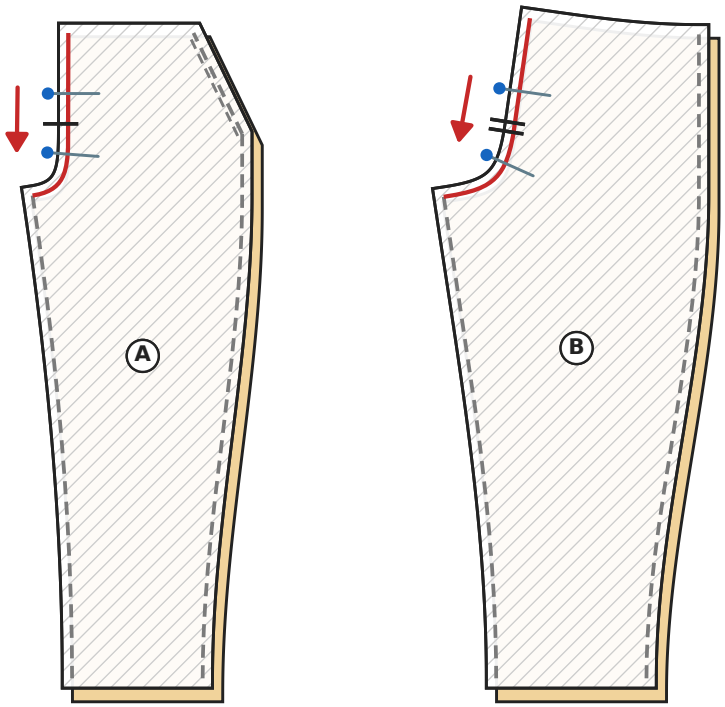



1. Turn leg 2 right side out.
2. Leg 1 stays wrong side out.
3. Push leg 2 into leg 1 until the top edges are even
4. The hems point the same way.

Tip: Now the legs lie right sides together and the crotch curves lie exactly on top of each other.

## 13 Sew the crotch seam

Sewing machine · Pins




1. Pin the crotch curves together, starting at the inseams: they lie exactly on top of each other.
2. Sew with 10 mm seam allowance in one go from the front top edge to the back top edge.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Sew the crotch curve a second time for strength.

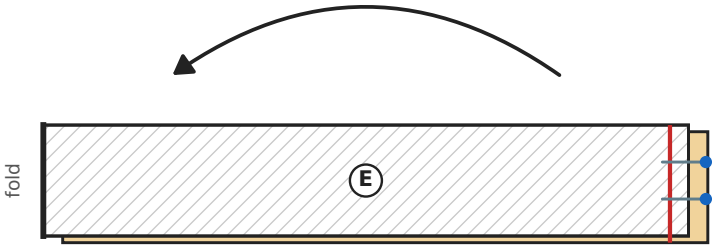
Tip: Press the inseam allowances in opposite directions: the crossing lies flat. Pull the inner leg out afterwards.

## 14 Close the waistband into a ring

Sewing machine · Pins

1. Fold the waistband crosswise right sides together and sew the short ends (centre back) with 10 mm seam allowance
2. Leave 3 cm open for the elastic.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

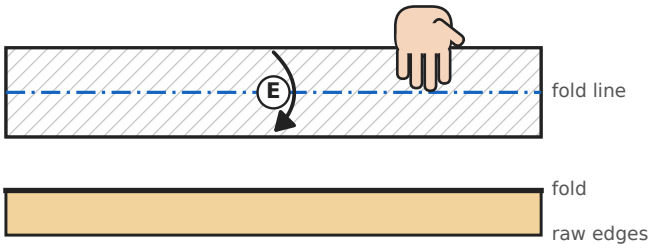


## 15 Fold the waistband in half

Iron

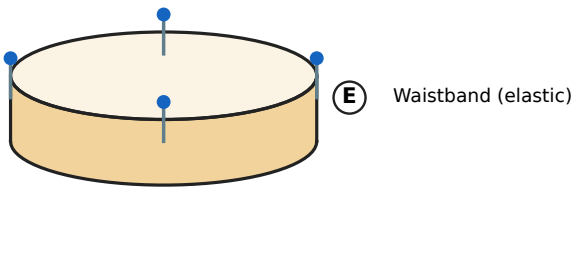
1. Fold lengthwise wrong sides together along the fold line and press.

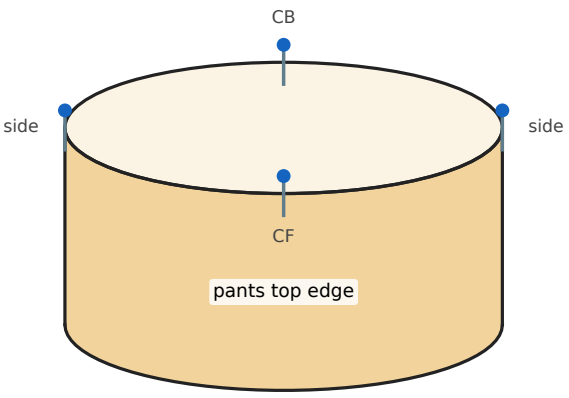
Iron: steam




## 16 Mark quarters

1. Mark centre front, centre back and both sides with pins, on the waistband and on the pants top edge.



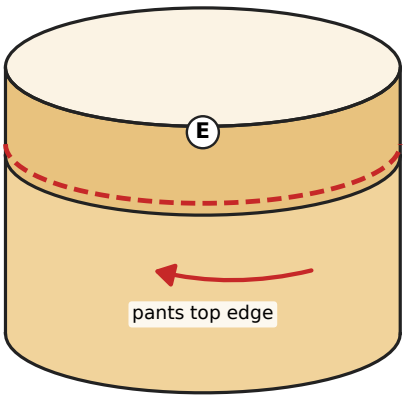



## 17 Sew the waistband on

Sewing machine · Pins

1. Pants right side out: slide the waistband over them, raw edges level, the opening at the back on the inside.
2. Match the quarter pins.
3. Sew all round with 10 mm seam allowance.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)



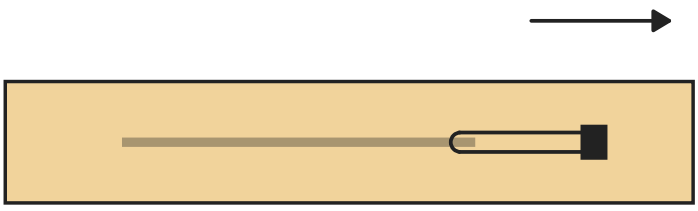

band folded, raw edges down

## 18 Thread the elastic

Sewing machine · Safety pin

1. With a safety pin through the opening
2. Overlap the ends by 2 cm and sew them together.

Sewing machine: wide zigzag 4 × 2.5 mm



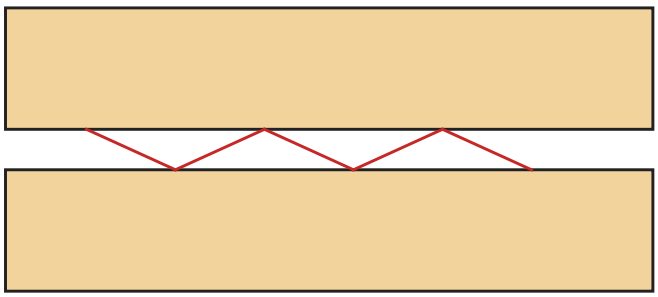
elastic, with a safety pin

## 19 Close the opening

Sewing machine

1. Spread the gathers evenly, close the opening and stitch in the ditch at the side seams so the elastic cannot twist.

Sewing machine: twin needle 4 mm, stitch 3 mm



ladder stitch: the stitches disappear

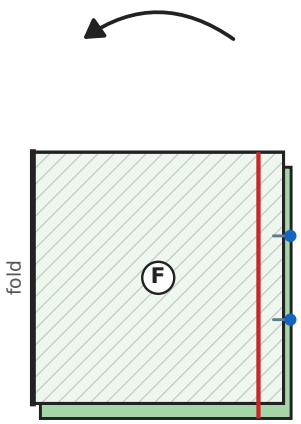
## 20 Close the cuffs into rings

Sewing machine

Pins

1. Fold each cuff crosswise right sides together, short edges level.
2. Sew with 10 mm seam allowance
3. Finger-press the allowances open.

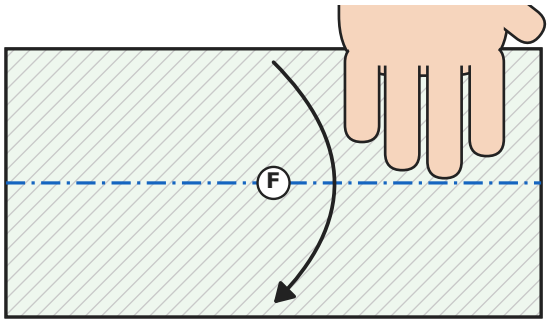
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)




2×

## 21 Fold the cuffs in half

Iron





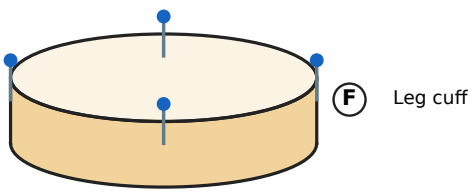


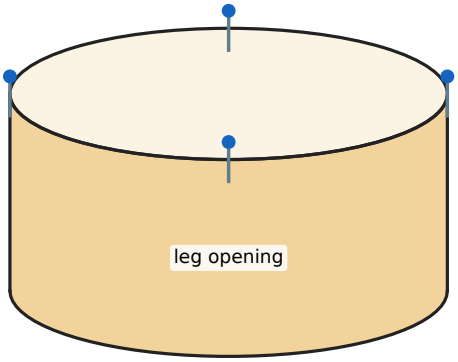
1. Fold the ring lengthwise wrong sides together, raw edges level.
2. The seam allowance is now inside
3. The right side shows all round.

Iron: steam

2×

## 22 Mark quarters



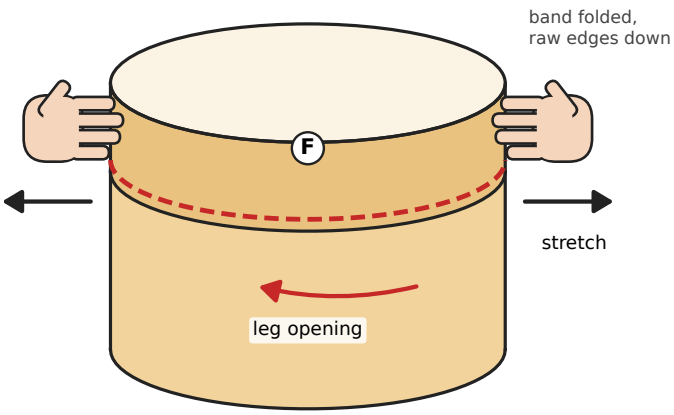

1. Mark quarters with pins on the cuff and on the leg opening
2. The cuff seam is one of them.

2×

## 23 Sew the cuffs on

Sewing machine

Pins




1. Turn the leg right side out.
2. Slide the cuff over the leg end, raw edges level, cuff seam on the inseam.
3. Match the quarter pins.
4. Sew with 10 mm seam allowance, cuff on top, stretching only the cuff.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

Tip: One hand in front of the presser foot, one behind it: both stretch evenly.

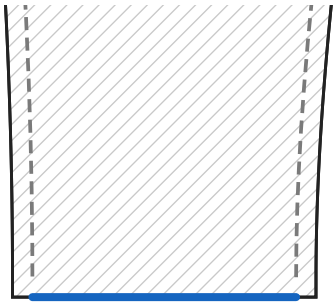
2×

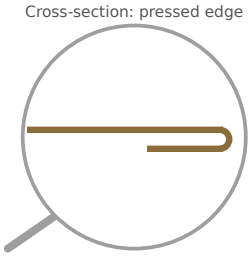
## 24 Press the allowance up

Iron

1. Press the seam allowance up towards the pants.

Iron: steam

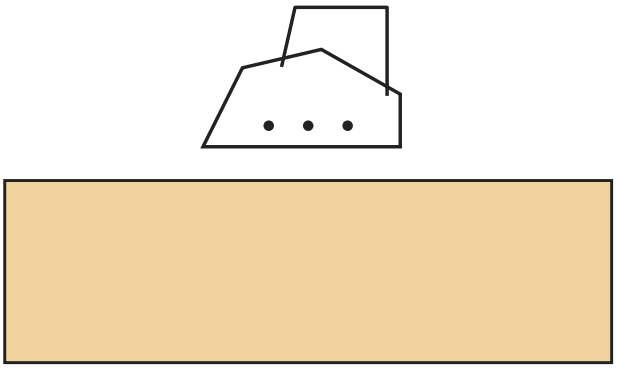




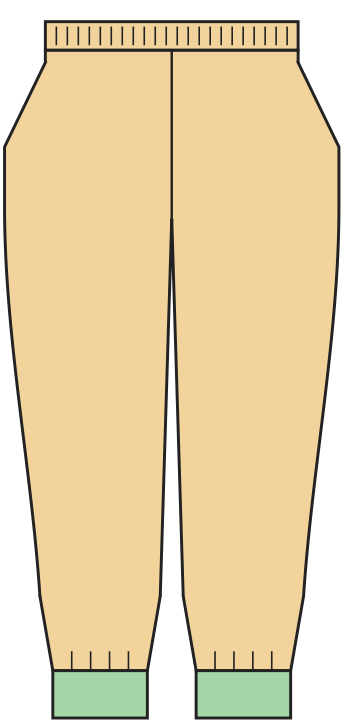

2×

## 25 Final pressing

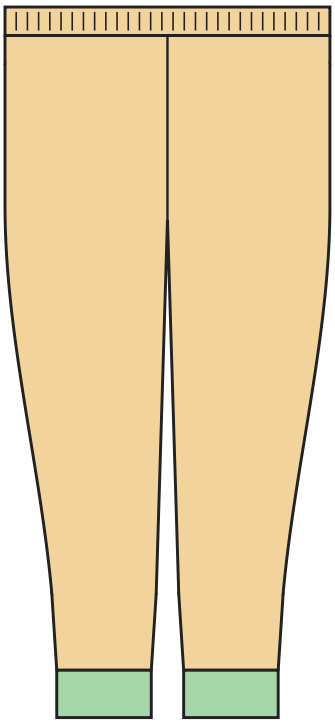
1. Press all seams with steam
2. Do not press down hard on sweat.



# Done!



Front view



Back view

After the test garment: Try the test garment on the child: length, width at the hip and the waistband. Adjust the measurements or options and print the pattern again before you cut the good fabric.

# Joggers · sizes 98 · 110 · 122

normal fit, Sweat / French Terry (25 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

## Files

- PDF roll 610 mm (600 × 1017 mm): `joggers_98_110_122_roll610.pdf`
- Preview roll 610 mm: `joggers_98_110_122_roll610_preview.png`
- SVG master 1:1: `joggers_98_110_122_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

## Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

## Body measurements

| Body measurement | Value | Source |
|---|---|---|
| Body height | 122.0 cm | table |
| Chest | 62.0 cm | table |
| Waist | 56.0 cm | table |
| Hip | 66.0 cm | table |
| Head | 52.0 cm | table |
| Neck | 28.5 cm | table |
| Inseam | 54.0 cm | table |
| Side length | 72.0 cm | table |
| Arm length | 47.0 cm | table |
| Back length | 29.5 cm | table |
| Shoulder width | 10.0 cm | table |
| Thigh | 37.0 cm | table |
| Ankle | 17.8 cm | table |

## Finished measurements (virtually sewn)

| Finished measurement | Value |
|---|---|
| Hip width | 72.6 cm |
| Thigh width | 46.4 cm |
| Leg width at mid-leg | 36.0 cm |
| Leg opening | 20.0 cm |
| Waistband relaxed | 50.4 cm |
| Waistband stretched (pants top edge) | 69.8 cm |
| Outseam | 73.5 cm |
| Inseam | 54.0 cm |
| Front rise | 19.5 cm |
| Back rise | 21.1 cm |
| Waistband height | 3.0 cm |
| Cuff height | 5.0 cm |
| Crotch seam total | 46.9 cm |

## Sizes in this pattern (nested)

All sizes are drawn on top of each other, aligned at the same construction point. Every size has its own colour and line and its own **PDF layer**: in Acrobat/Reader open the Layers panel and switch off the sizes you do not need before printing. Cut along the line of your size. Sizes can be blended: e.g. width of one size, length of another, by drawing a smooth line between the two outlines.

| Size | Line | Fabric needed | Hip width | Thigh width | Leg width at mid-leg | Leg opening |
|---|---|---|---|---|---|---|
| size 98 | red, solid | Main fabric: 0.65 m; Rib knit: 0.15 m | 62.7 cm | 39.9 cm | 31.3 cm | 17.2 cm |
| size 110 | blue, long dashes | Main fabric: 0.80 m; Rib knit: 0.15 m | 67.1 cm | 42.8 cm | 33.3 cm | 18.4 cm |
| size 122 | green, short dashes | Main fabric: 0.90 m; Rib knit: 0.20 m | 72.6 cm | 46.4 cm | 36.0 cm | 20.0 cm |

Fabric, finished measurements and sewing steps below refer to the largest size (size 122); the validation report has one table per size.

## Pattern pieces

| Piece | Cutting | Fabric | Seam allowance |
|---|---|---|---|
| Front leg | cut 2 (mirrored pair) | Main fabric | seam allowance 1.0 cm included |
| Back leg | cut 2 (mirrored pair) | Main fabric | seam allowance 1.0 cm included |
| Side panel (outer pocket bag) | cut 2 (mirrored pair) | Main fabric | seam allowance 1.0 cm included |
| Inner pocket bag | cut 2 (mirrored pair) | Main fabric or cotton jersey | seam allowance 1.0 cm included |
| Waistband (elastic) | cut 1 on fold | Main fabric | seam allowance 1.0 cm included |
| Leg cuff | cut 2 | Rib knit | seam allowance 1.0 cm included |

## Fabric needed

- **Main fabric:** about 0.90 m at 150 cm width (folded selvage to selvage, cut on fold pieces at the fold, incl. 10 % for shrinkage; single layer would need 0.90 m)
- **Rib knit:** about 0.20 m at 70 cm width (single layer, incl. 10 % for shrinkage)
- Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

**Cut on fold** (Waistband (elastic)): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. On the folded fabric, lay the fold edge exactly on the fabric fold, pin it and cut through both layers; opened out you get the whole piece. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

## Validation report

### Sewability check · size 98

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Seam pairs | Side seam front ↔ back | 51.4 cm | 51.4 cm | +0.0 mm | □ OK | |
| Seam pairs | Inseam front ↔ back | 35.9 cm | 35.9 cm | +0.0 mm | □ OK | |
| Seam pairs | Front crotch seam (left ↔ right piece) | 17.2 cm | 17.2 cm | +0.0 mm | □ OK | |
| Seam pairs | Back crotch seam (left ↔ right piece) | 20.8 cm | 20.8 cm | +0.0 mm | □ OK | |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Seam pairs | Pocket opening: front leg ↔ pocket bag | 10.5 cm | 10.5 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Pocket bag seam: pocket bag ↔ side panel | 19.4 cm | 19.4 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Pocket bag ↔ side panel at the side seam | 3.6 cm | 3.6 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Waistband ↔ pants top edge | 59.9 cm | 59.9 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Leg cuff ↔ leg opening (stretched) | 70–85 % of 24.6 cm | 17.2 cm = 70 % | -73.9 mm | ✓ OK | rib band is stretched while sewing |
| Angles/transitions | Centre front ↔ waistband | 90° | 90.0° | +0.0° | ✓ OK | |
| Angles/transitions | Centre back ↔ waistband | 90° | 90.0° | -0.0° | ✓ OK | |
| Angles/transitions | Crotch ↔ inseam (front + back) | 180° | 180.8° | +0.8° | ✓ OK | |
| Angles/transitions | Inseam ↔ hem front | 90° | 90.0° | +0.0° | ✓ OK | |
| Angles/transitions | Hem ↔ side seam front | 90° | 90.1° | +0.1° | ✓ OK | |
| Angles/transitions | Inseam ↔ hem back | 90° | 90.1° | +0.1° | ✓ OK | |
| Angles/transitions | Hem ↔ side seam back | 90° | 90.1° | +0.1° | ✓ OK | |
| Angles/transitions | Side seam ↔ waistband back | 90° | 90.1° | +0.1° | ✓ OK | |
| Angles/transitions | Side seam ↔ waistband at the side panel | 90° | 90.0° | +0.0° | ✓ OK | |
| Measurements | Hip circumference pattern = body + ease | 62.7 cm | 62.7 cm | +0.0 % | ✓ OK | body 57.0 cm + +10 % ease |
| Measurements | Top edge fits over the hip | ≥ 58.1 cm | 68.9 cm | +107.6 mm | ✓ OK | top edge incl. usable stretch ≥ hip |
| Measurements | Front rise plausible | 11.8 cm – 20.9 cm | 18.5 cm | +0.0 mm | ✓ OK | body rise 17.0 cm |
| Measurements | Inseam length plausible | 28.9 cm – 44.8 cm | 39.7 cm | +0.0 mm | ✓ OK | body inseam 40.0 cm |
| Measurements | Foot fits through the leg opening | ≥ 20.8 cm | 25.5 cm | +47.3 mm | ✓ OK | heel-instep circumference ≈ 1.35 × ankle |
| Geometry | Front leg: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Front leg: grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric |
| Geometry | Back leg: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Back leg: grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric |
| Geometry | Side panel (outer pocket bag): outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Side panel (outer pocket bag): grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Geometry | Inner pocket bag: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Inner pocket bag: grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric or cotton jersey |
| Geometry | Waistband (elastic): outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Waistband (elastic): grainline & cutting | present | cut 1 on fold | | ✓ OK | Main fabric |
| Geometry | Leg cuff: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Leg cuff: grainline & cutting | present | cut 2 | | ✓ OK | Rib knit |
| Geometry | Waistband (elastic): cut-on-fold edge | straight, no SA, 90° corners, on grain | ok, corners ≤ 0.0° off 90° | | ✓ OK | Centre front |
| Labels | Front leg: notches | on the cutting line | 3 notch mark(s) | | ✓ OK | |
| Labels | Back leg: notches | on the cutting line | 6 notch mark(s) | | ✓ OK | |

**Result:** 38 OK, 0 warning(s), 0 error(s): the pattern is sewable.

## Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 627}}`).

| Name | Value | Derived from | Note |
|---|---|---|---|
| `hip_width` | 62.7 cm | hip × (1 + 10 % ease) | |
| `front_back_offset` | 0.5 cm | front quarter narrower, back quarter wider by this amount | |
| `front_width` | 15.2 cm | hip_width / 4 − front_back_offset | |
| `back_width` | 16.2 cm | hip_width / 4 + front_back_offset | |
| `front_crotch_extension` | 3.1 cm | 5 % of hip_width | |
| `back_crotch_extension` | 6.3 cm | 10 % of hip_width | |
| `rise` | 15.5 cm | (side length − inseam) + seat ease − waistband height | |
| `inseam_finished` | 39.7 cm | inseam − allowance to the floor + blousing | |
| `hem_width` | 24.6 cm | cuff length / cuff stretch ratio | |

## Sewability check · size 110

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Seam pairs | Side seam front ↔ back | 59.0 cm | 59.0 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Inseam front ↔ back | 43.5 cm | 43.5 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Front crotch seam (left ↔ right piece) | 17.3 cm | 17.3 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Back crotch seam (left ↔ right piece) | 21.1 cm | 21.1 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Pocket opening: front leg ↔ pocket bag | 10.6 cm | 10.6 cm | +0.0 mm | ✓ OK | |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Seam pairs | Pocket bag seam: pocket bag ↔ side panel | 20.4 cm | 20.4 cm | +0.0 mm | □ OK | |
| Seam pairs | Pocket bag ↔ side panel at the side seam | 4.1 cm | 4.1 cm | +0.0 mm | □ OK | |
| Seam pairs | Waistband ↔ pants top edge | 64.3 cm | 64.3 cm | +0.0 mm | □ OK | |
| Seam pairs | Leg cuff ↔ leg opening (stretched) | 70–85 % of 26.4 cm | 18.4 cm = 70 % | -79.1 mm | □ OK | rib band is stretched while sewing |
| Angles/transitions | Centre front ↔ waistband | 90° | 90.0° | +0.0° | □ OK | |
| Angles/transitions | Centre back ↔ waistband | 90° | 90.0° | -0.0° | □ OK | |
| Angles/transitions | Crotch ↔ inseam (front + back) | 180° | 180.7° | +0.7° | □ OK | |
| Angles/transitions | Inseam ↔ hem front | 90° | 90.0° | +0.0° | □ OK | |
| Angles/transitions | Hem ↔ side seam front | 90° | 90.1° | +0.1° | □ OK | |
| Angles/transitions | Inseam ↔ hem back | 90° | 90.1° | +0.1° | □ OK | |
| Angles/transitions | Hem ↔ side seam back | 90° | 90.1° | +0.1° | □ OK | |
| Angles/transitions | Side seam ↔ waistband back | 90° | 90.1° | +0.1° | □ OK | |
| Angles/transitions | Side seam ↔ waistband at the side panel | 90° | 90.0° | +0.0° | □ OK | |
| Measurements | Hip circumference pattern = body + ease | 67.1 cm | 67.1 cm | +0.0 % | □ OK | body 61.0 cm + +10 % ease |
| Measurements | Top edge fits over the hip | ≥ 62.2 cm | 73.9 cm | +117.3 mm | □ OK | top edge incl. usable stretch ≥ hip |
| Measurements | Front rise plausible | 13.2 cm – 22.3 cm | 18.5 cm | +0.0 mm | □ OK | body rise 17.0 cm |
| Measurements | Inseam length plausible | 34.7 cm – 53.8 cm | 47.9 cm | +0.0 mm | □ OK | body inseam 48.0 cm |
| Measurements | Foot fits through the leg opening | ≥ 22.2 cm | 27.3 cm | +50.7 mm | □ OK | heel-instep circumference ≈ 1.35 × ankle |
| Geometry | Front leg: outline & seam allowance | closed, SA present | ok | | □ OK | seam allowance 1.0 cm included |
| Labels | Front leg: grainline & cutting | present | cut 2 (mirrored pair) | | □ OK | Main fabric |
| Geometry | Back leg: outline & seam allowance | closed, SA present | ok | | □ OK | seam allowance 1.0 cm included |
| Labels | Back leg: grainline & cutting | present | cut 2 (mirrored pair) | | □ OK | Main fabric |
| Geometry | Side panel (outer pocket bag): outline & seam allowance | closed, SA present | ok | | □ OK | seam allowance 1.0 cm included |
| Labels | Side panel (outer pocket bag): grainline & cutting | present | cut 2 (mirrored pair) | | □ OK | Main fabric |
| Geometry | Inner pocket bag: outline & seam allowance | closed, SA present | ok | | □ OK | seam allowance 1.0 cm included |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Labels | Inner pocket bag: grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric or cotton jersey |
| Geometry | Waistband (elastic): outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Waistband (elastic): grainline & cutting | present | cut 1 on fold | | ✓ OK | Main fabric |
| Geometry | Leg cuff: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Leg cuff: grainline & cutting | present | cut 2 | | ✓ OK | Rib knit |
| Geometry | Waistband (elastic): cut-on-fold edge | straight, no SA, 90° corners, on grain | ok, corners ≤ 0.0° off 90° | | ✓ OK | Centre front |
| Labels | Front leg: notches | on the cutting line | 3 notch mark(s) | | ✓ OK | |
| Labels | Back leg: notches | on the cutting line | 6 notch mark(s) | | ✓ OK | |

**Result:** 38 OK, 0 warning(s), 0 error(s): the pattern is sewable.

## Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 671}}`).

| Name | Value | Derived from | Note |
|---|---|---|---|
| `hip_width` | 67.1 cm | hip × (1 + 10 % ease) | |
| `front_back_offset` | 0.5 cm | front quarter narrower, back quarter wider by this amount | |
| `front_width` | 16.3 cm | hip_width / 4 − front_back_offset | |
| `back_width` | 17.3 cm | hip_width / 4 + front_back_offset | |
| `front_crotch_extension` | 3.4 cm | 5 % of hip_width | |
| `back_crotch_extension` | 6.7 cm | 10 % of hip_width | |
| `rise` | 15.5 cm | (side length − inseam) + seat ease − waistband height | |
| `inseam_finished` | 47.9 cm | inseam − allowance to the floor + blousing | |
| `hem_width` | 26.4 cm | cuff length / cuff stretch ratio | |

## Sewability check · size 122

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Seam pairs | Side seam front ↔ back | 65.7 cm | 65.7 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Inseam front ↔ back | 49.2 cm | 49.2 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Front crotch seam (left ↔ right piece) | 18.4 cm | 18.4 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Back crotch seam (left ↔ right piece) | 22.5 cm | 22.5 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Pocket opening: front leg ↔ pocket bag | 11.4 cm | 11.4 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Pocket bag seam: pocket bag ↔ side panel | 22.0 cm | 22.0 cm | +0.0 mm | ✓ OK | |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Seam pairs | Pocket bag ↔ side panel at the side seam | 4.5 cm | 4.5 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Waistband ↔ pants top edge | 69.8 cm | 69.8 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Leg cuff ↔ leg opening (stretched) | 70–85 % of 28.5 cm | 20.0 cm = 70 % | -85.5 mm | ✓ OK | rib band is stretched while sewing |
| Angles/transitions | Centre front ↔ waistband | 90° | 90.0° | +0.0° | ✓ OK | |
| Angles/transitions | Centre back ↔ waistband | 90° | 90.0° | -0.0° | ✓ OK | |
| Angles/transitions | Crotch ↔ inseam (front + back) | 180° | 180.7° | +0.7° | ✓ OK | |
| Angles/transitions | Inseam ↔ hem front | 90° | 90.0° | +0.0° | ✓ OK | |
| Angles/transitions | Hem ↔ side seam front | 90° | 90.1° | +0.1° | ✓ OK | |
| Angles/transitions | Inseam ↔ hem back | 90° | 90.0° | +0.0° | ✓ OK | |
| Angles/transitions | Hem ↔ side seam back | 90° | 90.1° | +0.1° | ✓ OK | |
| Angles/transitions | Side seam ↔ waistband back | 90° | 90.1° | +0.1° | ✓ OK | |
| Angles/transitions | Side seam ↔ waistband at the side panel | 90° | 90.0° | +0.0° | ✓ OK | |
| Measurements | Hip circumference pattern = body + ease | 72.6 cm | 72.6 cm | +0.0 % | ✓ OK | body 66.0 cm + +10 % ease |
| Measurements | Top edge fits over the hip | ≥ 67.3 cm | 80.3 cm | +129.4 mm | ✓ OK | top edge incl. usable stretch ≥ hip |
| Measurements | Front rise plausible | 14.6 cm – 24.0 cm | 19.5 cm | +0.0 mm | ✓ OK | body rise 18.0 cm |
| Measurements | Inseam length plausible | 39.0 cm – 60.5 cm | 54.0 cm | +0.0 mm | ✓ OK | body inseam 54.0 cm |
| Measurements | Foot fits through the leg opening | ≥ 24.1 cm | 29.5 cm | +54.8 mm | ✓ OK | heel-instep circumference ≈ 1.35 × ankle |
| Geometry | Front leg: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Front leg: grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric |
| Geometry | Back leg: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Back leg: grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric |
| Geometry | Side panel (outer pocket bag): outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Side panel (outer pocket bag): grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric |
| Geometry | Inner pocket bag: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Inner pocket bag: grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric or cotton jersey |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Geometry | Waistband (elastic): outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Waistband (elastic): grainline & cutting | present | cut 1 on fold | | ✓ OK | Main fabric |
| Geometry | Leg cuff: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Leg cuff: grainline & cutting | present | cut 2 | | ✓ OK | Rib knit |
| Geometry | Waistband (elastic): cut-on-fold edge | straight, no SA, 90° corners, on grain | ok, corners ≤ 0.0° off 90° | | ✓ OK | Centre front |
| Labels | Front leg: notches | on the cutting line | 3 notch mark(s) | | ✓ OK | |
| Labels | Back leg: notches | on the cutting line | 6 notch mark(s) | | ✓ OK | |

**Result:** 38 OK, 0 warning(s), 0 error(s): the pattern is sewable.

## Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 726}}`).

| Name | Value | Derived from | Note |
|---|---|---|---|
| `hip_width` | 72.6 cm | hip × (1 + 10 % ease) | |
| `front_back_offset` | 0.5 cm | front quarter narrower, back quarter wider by this amount | |
| `front_width` | 17.7 cm | hip_width / 4 − front_back_offset | |
| `back_width` | 18.7 cm | hip_width / 4 + front_back_offset | |
| `front_crotch_extension` | 3.6 cm | 5 % of hip_width | |
| `back_crotch_extension` | 7.3 cm | 10 % of hip_width | |
| `rise` | 16.5 cm | (side length − inseam) + seat ease − waistband height | |
| `inseam_finished` | 54.0 cm | inseam − allowance to the floor + blousing | |
| `hem_width` | 28.5 cm | cuff length / cuff stretch ratio | |

